

# **CSE113**

# **Foundations of Computer Science**

**State University of New York, Korea**

# Previous lectures

Truth table

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

Equivalence laws

| Laws              | Formula                                                     | Formula                                                   |
|-------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| Commutative laws  | $p \wedge q \equiv q \wedge p$                              | $p \vee q \equiv q \vee p$                                |
| Associative laws  | $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$        | $(p \vee q) \vee r \equiv p \vee (q \vee r)$              |
| Distributive laws | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ |

**Propositional logic: A formal language to express facts and argue about them**

Valid arguments

|                         |
|-------------------------|
| Premise <sub>1</sub>    |
| Premise <sub>2</sub>    |
| ⋮                       |
| Premise <sub>m</sub>    |
| $\therefore$ Conclusion |

Inference

| Name          | Rule                                                 | Name         | Rule                                                                        |
|---------------|------------------------------------------------------|--------------|-----------------------------------------------------------------------------|
| Modus Ponens  | $p \rightarrow q$<br>$p$<br>$\therefore q$           | Elimination  | $p \vee q$ $p \vee q$<br>$\sim q$ $\sim p$<br>$\therefore p$ $\therefore q$ |
| Modus Tollens | $p \rightarrow q$<br>$\sim q$<br>$\therefore \sim p$ | Transitivity | $p \rightarrow q$<br>$q \rightarrow r$<br>$\therefore p \rightarrow r$      |

# Plan

- Revision on using Inference rules for validity
- Predicate Logic, or propositions with Quantifiers
- Negation

# **Revision on Using Inference Rules for Validity**

In 41–44 a set of premises and a conclusion are given. Use the valid argument forms listed in Table 2.3.1 to deduce the conclusion from the premises, giving a reason for each step as in Example 2.3.8. Assume all variables are statement variables.

- 41.** a.  $\sim p \vee q \rightarrow r$   
b.  $s \vee \sim q$   
c.  $\sim t$   
d.  $p \rightarrow t$   
e.  $\sim p \wedge r \rightarrow \sim s$   
f.  $\therefore \sim q$

42. a.  $p \vee q$   
b.  $q \rightarrow r$   
c.  $p \wedge s \rightarrow t$   
d.  $\sim r$   
e.  $\sim q \rightarrow u \wedge s$   
f.  $\therefore t$

# Propositions with Quantifiers



# Why quantifiers? To express “for all” and “there exists”

- Everyone can make mistake
- Nobody is perfect
- Every lock has a key
- There is a key for every lock
- Every nonzero real number has a reciprocal



# CS example 1:

## Software security

Question: Could the program print “bad”?

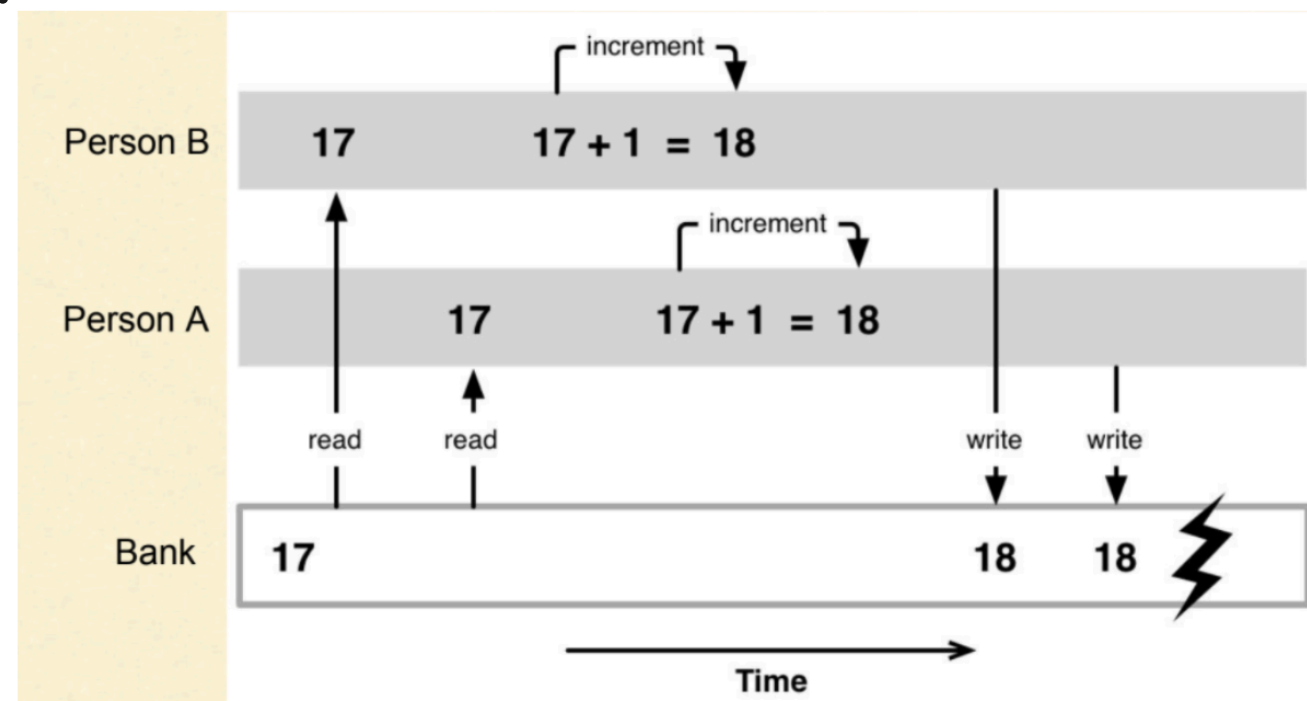
```
1  #include <stdio.h>
2
3  void f(int input) {
4      char a[8];
5      int b = 0;
6      a[input] = 1;
7
8      if (b == 0)
9          printf("good\n");
10     else
11         printf("bad\n");
12 }
```

**Exisitential Statement**

: there exists an integer  $n$ , such that  $b \neq 0$  at line 8 when executing  $f(n)$

# CS example 2: Concurrency

Two persons are trying to deposit 1 dollar online into the same bank account.



**Universal Statement:**

**For all CPU schedule  $s$ , A should not read while B intends to write.**

# Predicate

- A **propositional function** or **predicate** is a sentence that contains **one or more variables**
- A predicate is neither true nor false
- A predicate becomes a proposition when the variables are substituted with specific values
- The **domain** of a predicate variable is the set of all values that may be substituted for the variable

## Examples

| Symbol    | Predicate      | Domain                               | Propositions                 |
|-----------|----------------|--------------------------------------|------------------------------|
| $p(x)$    | $x > 5$        | $x \in \mathbb{R}$                   | $p(6), p(-3.6), p(0), \dots$ |
| $p(x, y)$ | $x + y$ is odd | $x \in \mathbb{Z}, y \in \mathbb{Z}$ | $p(4, 5), p(-4, -4), \dots$  |

# Universal statement

- Let  $p(x)$  be a predicate and  $D$  be the domain of  $x$
- A **universal statement** is a statement of the form

$$\forall x \in D, p(x)$$

- Forms:
  - “ $p(x)$  is true for all values of  $x$ ”
  - “For all  $x$ ,  $p(x)$ ”
  - “For each  $x$ ,  $p(x)$ ”
  - “For every  $x$ ,  $p(x)$ ”
  - “Given any  $x$ ,  $p(x)$ ”

# Existential statement

- Let  $p(x)$  be a predicate and  $D$  be the domain of  $x$
- An **existential statement** is a statement of the form

$$\exists x \in D, p(x)$$

- Forms:
  - “There exists an  $x$  such that  $p(x)$ ”
  - “For some  $x$ ,  $p(x)$ ”
  - “We can find an  $x$ , such that  $p(x)$ ”
  - “There is some  $x$  such that  $p(x)$ ”
  - “There is at least one  $x$  such that  $p(x)$ ”

Examples

| Universal st.s                         | Domain            |
|----------------------------------------|-------------------|
| $\exists x \in D, x^2 \geq x$          | $D = \{1, 2, 3\}$ |
| $\exists x \in \mathbb{R}, x^2 \geq x$ | $\mathbb{R}$      |

# Propositions with multiple quantifiers

$\forall x \in D, \exists y \in E$ , such that  $p(x, y)$

Every lock has a key.

# Propositions with multiple quantifiers

$$\exists x \in D, \forall y \in E, \text{ such that } p(x, y)$$

There is a key for every lock.



# Propositions with multiple quantifiers

$\forall x \in D, \forall y \in E$ , such that  $p(x, y)$

“Give me a place to stand, and a lever long enough, and I will move the Earth.”



# Propositions with multiple quantifiers

$\exists x \in D, \exists y \in E$ , such that  $p(x, y)$

“There is someone in a park sitting on a bench”.

# Four Notes

# Note 1:

## the order of quantifiers matters

- Every lock has a key
- For any lock  $L$ , there exists a key  $K$ , such that  $K$  can unlock  $L$ .
- There is a key for every lock
- There exists a key  $K$ , such that for any lock  $L$ ,  $K$  can unlock  $L$ .

# Note 2: A commonly used notational equivalence

$$\forall x \in D, p(x)$$

Equivalent to

$$\forall x, x \in D \rightarrow p(x)$$

# Example: All doctors wear glasses

- for all  $d$ , if  $d$  is a doctor, then  $d$  wears glasses
- Formally, if we define
  - $D$  to be the set of doctors,
  - $\text{wear\_glass}$  to be a function that takes a person  $x$  as an input, and returns true if  $x$  wears glasses
- then the following two statements are considered the same

$$\forall d \in D, \text{wear\_glass}(d).$$

$$\forall d, d \in D \rightarrow \text{wear\_glass}(d)$$

# Note 3: Universal conditional statement

- A **universal conditional statement** is of the form

$$\forall x, \text{ if } p(x) \text{ then } q(x)$$

## Examples

- $\forall x \in \mathbb{R}, \text{ if } x > 2 \text{ then } x^2 > 4$
- $\forall$  real number  $x$ , if  $x$  is an integer then  $x$  is rational  
 $\forall$  integer  $x$ ,  $x$  is rational
- $\forall x$ , if  $x$  is a square then  $x$  is a rectangle  
 $\forall$  square  $x$ ,  $x$  is a rectangle

# Note 4: Implicit quantifiers

## Examples

- If **a number** is an integer, then it is a rational number  
Implicit meaning:  $\forall$  number  $x$ , if  $x$  is an integer,  $x$  is rational
- **The number** 10 can be written as a sum of two prime numbers  
Implicit meaning:  $\exists$  prime numbers  $p$  and  $q$  such that  $10 = p + q$
- If  $x > 2$ , then  $x^2 > 4$   
Implicit meaning:  $\forall$  real  $x$ , if  $x > 2$ , then  $x^2 > 4$

## Definition

- Let  $p(x)$  and  $q(x)$  be predicates and  $D$  be the common domain of  $x$ . Then implicit quant. symbols  $\Rightarrow, \Leftrightarrow$  are defined as:

$$p(x) \Rightarrow q(x) \equiv \forall x, p(x) \rightarrow q(x)$$

$$p(x) \Leftrightarrow q(x) \equiv \forall x, p(x) \leftrightarrow q(x)$$



# Exercises



# **Exercise 1**

## **translate to formal logic**

**Everyone can make mistake**

# Exercise 2

## translate to formal logic

Nobody is perfect

# Exercise 3

## translate to formal logic

Every nonzero real number has a reciprocal

The reciprocal of 4 is  $\frac{1}{4}$  (namely 0.25)

# Exercise 4: Translate to formal logic

- Given:
  - $S$ : set of students
  - $P(s)$ :  $s$  passed the exam.
  - $W(s)$ :  $s$  worked hard.
  - $C(s)$ :  $s$  is in Professor Cho's class.
- **Every student in Professor Cho's class passed the exam**

# Exercise 5: Translate to formal logic

- Given:
  - $S$ : set of students
  - $P(s)$ :  $s$  passed the exam.
  - $W(s)$ :  $s$  worked hard.
  - $C(s)$ :  $s$  is in Professor Cho's class.
- **Some students studied hard but did not pass the exam**

# Exercise 6: Translate to formal logic

- Given:
  - $S$ : set of students
  - $P(s)$ :  $s$  passed the exam.
  - $W(s)$ :  $s$  worked hard.
  - $C(s)$ :  $s$  is in Professor Cho's class.
- **There are students who did not study hard but passed the exam**

# Exercise 7: Translate to formal logic

- Given:
  - $S$ : set of students
  - $P(s)$ :  $s$  passed the exam.
  - $W(s)$ :  $s$  worked hard.
  - $C(s)$ :  $s$  is in Professor Cho's class.
- **All students who studied hard passed the exam.**



# Exercise 8: Translate to formal logic

- Given:
  - $S$ : set of students
  - $P(s)$ :  $s$  passed the exam.
  - $W(s)$ :  $s$  worked hard.
  - $C(s)$ :  $s$  is in Professor Cho's class.
- **No student in Professor Cho's class failed the exam.**



# Exercise 9: Translate to formal logic

- Given:
  - L: set of locks
  - K set of keys
  - unlock (k,l): k can unlock l
- **Some keys cannot unlock any lock.**

# Negation

# Negation of a universal statement

$$\sim (\forall x \in D, p(x)) \equiv \exists x \in D, \sim p(x)$$

Example: "All swans are white."

Negation: "There exists at least one swan that is not white."

# Exercise

- Negate "Every phone on the table is turned off."

# Negation of an existential statement

$$\sim (\exists x \in D, p(x)) \equiv \forall x \in D, \sim p(x)$$

Example: "There is a car in the parking lot that is electric."

Negation: "No car in the parking lot is electric", or "For every car  $c$  in the parking lot,  $c$  is not electric"

# Exercise

- Negate "There is a person in the village who speaks Italian."

# Summary

$$\sim (\forall x \in D, p(x)) \equiv \exists x \in D, \sim p(x)$$

$$\sim (\exists x \in D, p(x)) \equiv \forall x \in D, \sim p(x)$$

- Negation of a **universal** statement (“all are”) is logically equivalent to an **existential** statement (“there is at least one that is not”)

Negation of an **existential** statement (“some are”) is logically equivalent to a **universal** statement (“all are not”)



# Exercise

- $\forall$  primes  $p$ ,  $p$  is odd



# Exercise

- $\exists$  triangle  $T$ , sum of angles of  $T$  equals  $200^\circ$

# Exercise

- No child is left behind

# Common mistakes

## Examples

- All mathematicians wear glasses

Negation (**incorrect**): No mathematician wears glasses

Negation (**correct**): There is at least one mathematician who does not wear glasses

- Some snowflakes are the same

Negation (**incorrect**): Some snowflakes are different

Negation (**correct**): All snowflakes are different

# Negation of universal conditional statements

## Definition

- Formally,

$$\begin{aligned}\sim (\forall x, p(x) \rightarrow q(x)) &\equiv \exists x, \sim (p(x) \rightarrow q(x)) \\ &\equiv \exists x, (p(x) \wedge \sim q(x))\end{aligned}$$

# Negation of universal conditional statements

## Definition

- Formally,

$$\begin{aligned}\sim (\forall x, p(x) \rightarrow q(x)) &\equiv \exists x, \sim (p(x) \rightarrow q(x)) \\ &\equiv \exists x, (p(x) \wedge \sim q(x))\end{aligned}$$

## Examples

- $\forall$  real  $x$ , if  $x > 10$ , then  $x^2 > 100$ .  
Negation:  $\exists$  real  $x$  such that  $x > 10$  and  $x^2 \leq 100$ .
- If a computer program has more than 100,000 lines, then it contains a bug.  
Negation: There is at least one computer program that has more than 100,000 lines and does not contain a bug.

# Negation of propositions with multiple quantifiers

$$\begin{aligned} & \sim (\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)) \\ & \equiv \exists x \text{ in } D \text{ such that } \sim (\exists y \text{ in } E \text{ such that } P(x, y)) \\ & \equiv \exists x \text{ in } D \text{ such that } \forall y \text{ in } E, \sim P(x, y) \end{aligned}$$

$$\begin{aligned} & \sim (\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)) \\ & \equiv \forall x \text{ in } D, \sim (\forall y \text{ in } E, P(x, y)) \\ & \equiv \forall x \text{ in } D, \exists y \text{ in } E \text{ such that } \sim P(x, y) \end{aligned}$$

# Exercise

- Do some research: Formal definition of continuity of a real-valued function  $f$  on a point  $x$
- Give a formal definition of  $f$  being discontinuous on  $x$



# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a person.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **All people are busy.**



# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a person.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **Some people are not busy.**

# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a person.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **Every person likes themselves.**

# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a person.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **There's someone who doesn't like themselves.**

# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a person.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **There's at least one person that everyone likes.**

# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a person.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **Everyone likes at least one person.**

# Exercise: Translate to formal logic, and then negate

- Given:
  - $P(x)$ :  $x$  is a doctor.
  - $Q(x)$ :  $x$  is busy.
  - $R(x, y)$ :  $x$  likes  $y$ .
- **Some people don't like themselves.**